

Candi Zheng

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Education

- **Ph.D. in Mathematics**, Feb 2019 – Present, Graduation date by Aug 2024
The Hong Kong University of Science and Technology (Supervisor: Professor [Yang Wang](#))
- **Ph.D. in Mechanical and Aerospace Engineering**, Sep 2017 – Feb 2019
The Hong Kong University of Science and Technology (Supervisor: Professor [Yang Wang](#))
- **Joint Ph.D. in Mechanical and Aerospace Engineering**, Feb 2017 – Present
The Southern University of Science and Technology (Supervisor: Professor [Shiyi Chen](#))
- **Visiting Student**, Sep 2016 – Jan 2017
The University of Edinburgh
- **B.S. in Physics**, Sep 2013 – Jun 2017
The Southern University of Science and Technology, (Supervisor: Professor [Hu Xu](#))

Research Interest

Science4AI and AI4Science: My research focuses on AI and non-equilibrium physics, particularly generative modeling and kinetic theory, with an emphasis on physics insights and mathematical solidity.

1. Generative Diffusion Model: How does physics knowledge enhance image generation quality and control in diffusion models?
2. Non-Equilibrium Physics: how does AI help modeling better constitutive relations for kinetic theory, rarefied gas dynamics, shock waves, and light scattering?

Publications

1. **Candi Zheng**^{*†}, Yuan Lan[†]. “[Characteristic Guidance: Non-linear Correction for DDPM at Large Guidance Scale](#).” ICML 2024.
2. **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “[Phase Transition in Extended Thermodynamics Triggers Sub-shocks](#).” arXiv:2304.10742 (2023).
3. **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “[Stabilizing the Maximal Entropy Moment Method for Rarefied Gas Dynamics at Single-Precision](#).” arXiv:2303.02898 (2023).
4. **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “[Data-driven constitutive relation reveals scaling law for hydrodynamic transport coefficients](#).” Physical Review E 107, no. 1 (2023): 015104.

5. Jin, Yuan-Jun, Rui Wang, Jin-Zhu Zhao, Yong-Ping Du, **Can-Di Zheng**, Li-Yong Gan, Jun-Feng Liu, Hu Xu*, and S. Y. Tong*. “[The prediction of a family group of two-dimensional node-line semimetals](#).” *Nanoscale* 9, no. 35 (2017): 13112-13118.

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Projects

1. [Characteristic Guidance Web UI](#) is an extension of for the Stable Diffusion web UI (AUTOMATIC1111). It offers a theory-backed guidance sampling method with improved sample and control quality at high CFG scale.
2. [Moment Gauge](#) is a Python JAX library designed to facilitate the implementation of numerical solvers using the maximal entropy moment method.

Awards and Scholarships

- **Risk Classification Top 10**, ATEC 2018 Developer AI Challenge, Primary round, 2018
- **Outstanding Graduation Project**, Southern University of Science and Technology, 2017
- **The University Physics Competition**, Silver, 2015
- **RoboMaster**, Central South China Division, Third Class, 2015
- **National Encouragement Scholarship**, Southern University of Science and Technology, 2014
- **Outstanding Freshman Scholarship**, Southern University of Science and Technology, 2014

Skills

- **Language:** English, Chinese, A little bit of Japanese
- **Math:** Ph.D. qualification in pure mathematics options (analysis and algebra).
- **Physics:** Specialized in statistical physics and kinetic theory.
- **Programming:** Proficient in Python for a diverse range of applications. Skilled at object oriented programming (OOP). Experience with JAVA, Fortran, CUDA C, and MPI for high-performance computing tasks.
- **Machine Learning:** Proficient in common machine learning frameworks and tools, including Pytorch, Jax, and Scikit-learn.